

X02 INTERIOR HAZARDOUS MATERIALS MANAGEMENT | P

Intent : Manage risks of human exposure to hazardous materials ubiquitously used in past construction practices.

Summary : This WELL feature requires the application of practices to manage exposure risks of the hazardous building materials asbestos, lead and polychlorinated biphenyls (PCBs).

Issue : Some materials that were commonly used in construction due to their mechanical, chemical or visual properties has shown to be toxic, and are now banned or confined to limited production across the world. However, their ubiquitous presence still haunts our built and natural environments and can cause disease upon exposure. Asbestos, lead and polychlorinated biphenyls (PCBs)-containing materials are emblematic examples. Asbestos, a naturally present, chemically resistant material found in old adhesives, insulation and sheeting is a known toxicant and carcinogen. Exposure to asbestos-containing dusts is the main cause of mesothelioma and responsible of over 200,000 deaths annually.¹ Lead is found in paints for increasing their durability and impermeability. If ingested, absorbed or breathed-in, it accumulates in blood, tissues and bones, potentially disrupting body functions and impairing the intellectual development of children and the unborn during pregnancy.² PCBs were used in caulk and electrical equipment, but due to their persistence in the environment, bioaccumulation in foods and carcinogenicity, their production is banned worldwide.³

Solutions : Existing buildings must be evaluated for the presence of hazardous materials and consider their removal when technically feasible. If not possible, hazards must be isolated and periodically monitored to prevent human exposure. Generating dust that can become suspended and respirable must be avoided. Any waste generated from construction, or any other onsite activity must be handled appropriately in accordance with best practices.

PART 1 MANAGE ASBESTOS HAZARDS

For All Spaces:

Option 1: Asbestos risk assessment and remediation

The following requirements are met:

- a. The building was constructed or last renovated before the enactment of laws banning the installation of asbestos-containing materials, or is located where there is no local asbestos phase-out regulation.
- b. An inspector certified under local regulation or a qualified professional with demonstrable experience where no local regulations apply conducts an investigation of the project space and reports the following:
 1. A list of locations where presumed asbestos containing materials (PACM) were found.
 2. Confirmation of the presence of asbestos is performed through Polarized Light Microscopy (PLM), Scanning Electron Microscopy (SEM) or Transmission Electron Microscopy (TEM) testing. The sample number and location follow applicable laws or recommendations of the inspector conducting the assessment. Materials having over 1% of asbestos are considered ACM. If analytical confirmation is not available or possible, all PACM are considered asbestos-containing materials (ACM).
- c. If asbestos-containing materials (ACM) were found per the above, an action plan that contains the following is implemented:
 1. Notification of any works to relevant authorities and persons living, working or transiting in the vicinity of the building or space.
 2. Preventative measures against the formation and spread of asbestos fibers in the air during remedial work.
 3. Measures taken for workers' protection during remediation activities, including but not limited to skin and respiratory protection.
 4. If ACM are being removed, activities are carried out for proper handling of ACM waste, including: wetting of all removed ACM, care in transportation to prevent crumbling, sealing and leak-tight transportation, proper labeling and final disposal in locations allowed by applicable laws and permits.
 5. Post-remediation clearance for occupancy confirmation by testing of fibers in air using phase contrast microscopy (PCM) or transmission electron microscopy (TEM) following standards referenced in applicable local laws or, if not available, NIOSH Manual of Analytical Methods (MNAM) Methods 7400 or 7402, GBZ/T192.5-2007, ISO 8672:2014, ISO 10312:2019 or ISO 13794:2019. The number of samples and sampling conditions must meet local regulations and/or conform to ISO 16000-7.
 6. If any of the asbestos is managed by methods other than removal, the month and year of follow-up inspection to evaluate the structural integrity of the ACM must be stated and cannot exceed three years from the date of the last inspection.

OR

Option 2: New spaces

The following requirement is met:

- a. The building was constructed after the enactment of an asbestos ban in construction products.

OR

Option 3: Demonstration of prior remediation

The following requirements are met:

- a. The project was last renovated after the enactment of local laws banning the installation of asbestos-containing materials.

- b. The project demonstrates that asbestos remediation and clearance is a legal requirement to grant occupancy of the space.

PART 2 MANAGE LEAD PAINT HAZARDS

For All Spaces:

Option 1: Identify lead paint hazards

The following requirements are met:

- a. A certified inspector or a qualified professional where no local regulations apply conducts an investigation of the space and reports the following:
 - 1. An inventory of locations of potential sources and sinks of lead-containing materials, where lead-containing paint may be present.
 - 2. Confirmation of lead hazards through in-situ test results by x-ray fluorescence (XRF) or by laboratory analyses of paint chips⁴ and/or surface dusts. Surface dust is considered a hazard if its lead loading is more than 0.11 mg/m² of the collection area if sampled from floors or over 1.08 mg/m² for dust on interior window sills.⁵ Paints having over 0.5% of lead by weight or 10,000 mg/m² of applied area and bare soil containing over 400 ppm of lead by weight are also considered lead hazards.⁵ Lower thresholds mandated by local regulations prevail for terms of hazard assessment.
- b. If lead is found in the investigation, a certified inspector (or a qualified professional where no local regulations apply) implements an action plan that contains the following:
 - 1. Notification of remediation work to occupants and transient populations in the surrounding spaces, and restriction of access to work areas during remediation.
 - 2. If paints are mechanically removed, measures are taken to minimize the formation and spread of dusts during the remediation process and to ensure adequate respiratory and skin protection for workers.
 - 3. A re-inspection schedule that includes visual assessments and dust testing, if any lead-containing paints are left in place and are subject to stabilization (i.e., painted over with products to prevent chipping or degradation) or enclosure, at least once every three years.
 - 4. Post-remediation clearance, confirming that the lead loading in dust is below the levels deemed hazardous.

OR

Option 2: New spaces

The following requirement is met:

- a. The building was constructed after the enactment of lead paint ban.

OR

Option 3: Demonstration of prior remediation

The following requirements are met:

- a. The project was last renovated after the enactment of local laws banning the application of lead-containing paint.
- b. The project demonstrates through legal documentation (e.g., approved certificates of occupancy, remediation reports submitted to relevant authorities) that lead remediation was performed and clearance was provided.

PART 3 MANAGE POLYCHLORINATED BIPHENYL (PCB) HAZARDS

For All Spaces:

Option 1: PCB remediation

The following requirements are met:

- a. The building was constructed or last renovated before the institution of any applicable laws banning or restricting PCBs, and is undergoing renovation work that disturbs (i.e., partially or fully removes) materials likely to contain PCBs such as caulking, fluorescent light ballasts and capacitors of appliances fabricated before 1980.
- b. An inspection strategy for assessing PCB-related risks is implemented and contains the following:
 - 1. Determination of locations where materials potentially containing PCBs may be disturbed.
 - 2. If caulk is to be disturbed or removed, analysis of the presumably PCB-containing material following protocols mandated by local laws or, in absence of local laws, by any applicable US EPA⁶ or ISO testing methods.
- c. If PCBs are found in disturbed materials, an action plan is implemented and contains the following:
 - 1. Notification of remedial work to relevant authorities and building occupants.
 - 2. Preventative measures against the spread of PCB-containing dusts and human exposure during remediation activities, including restricting access for those not involved in the work.

3. Protective measures for workers, including chemical-resistant gloves, clothing protection, goggles and respirators.
4. Waste handling that minimizes the spread of contaminated debris and safe disposal of PCB-containing waste in locations allowed by applicable local regulations.

OR

Option 2: No PCB remediation

One of the following is met:

- a. The building was constructed or last renovated before the institution of any applicable laws banning or restricting PCBs, and the project is not undergoing renovation work that disturbs (i.e., partially or fully removes) materials likely to contain PCBs such as caulking, fluorescent light ballasts and capacitors of appliances fabricated before 1980.
- b. The building was constructed or last renovated after the institution of any applicable laws banning or restricting PCBs.

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